

8. Ticklish Little Guy

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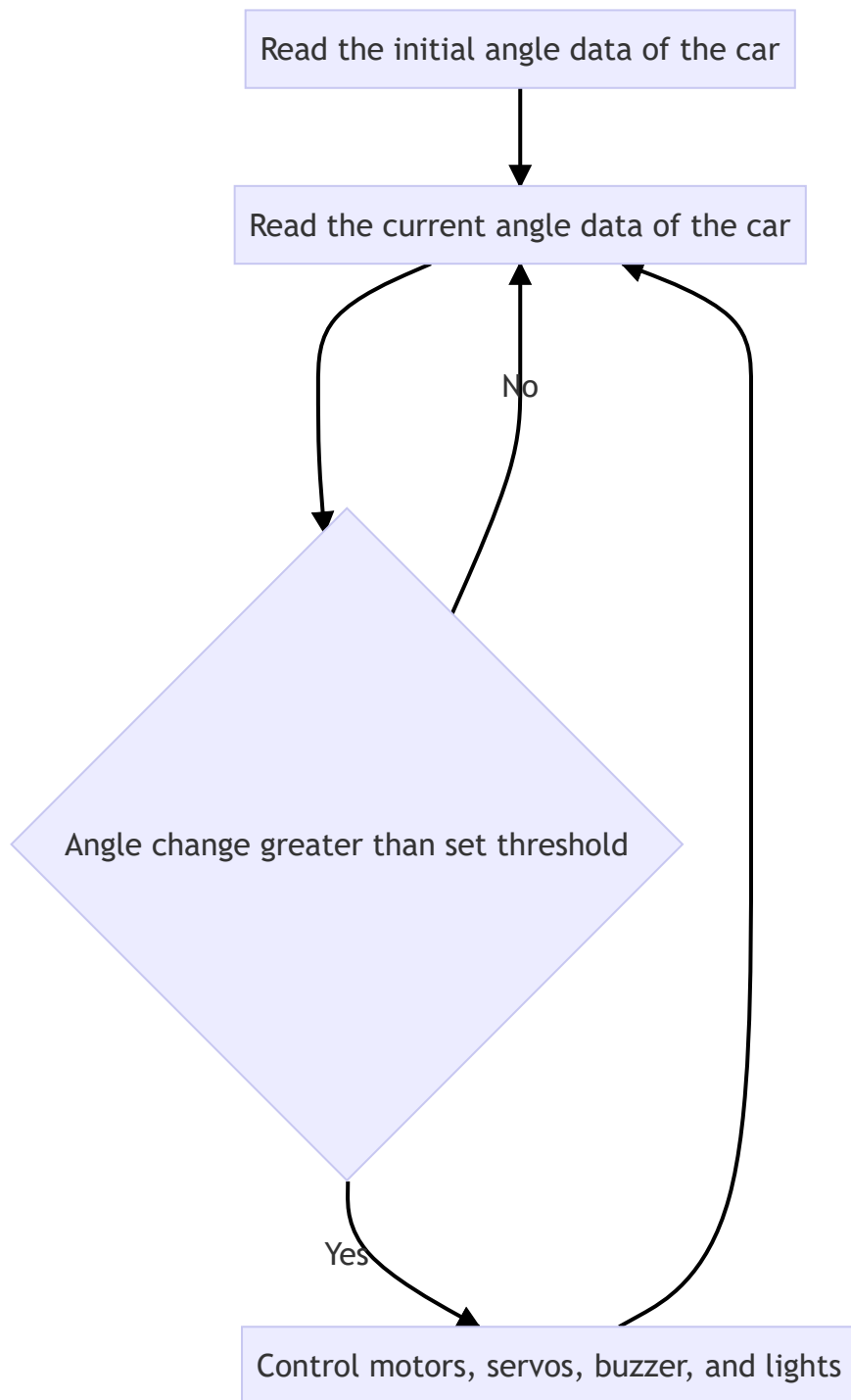
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Function Description

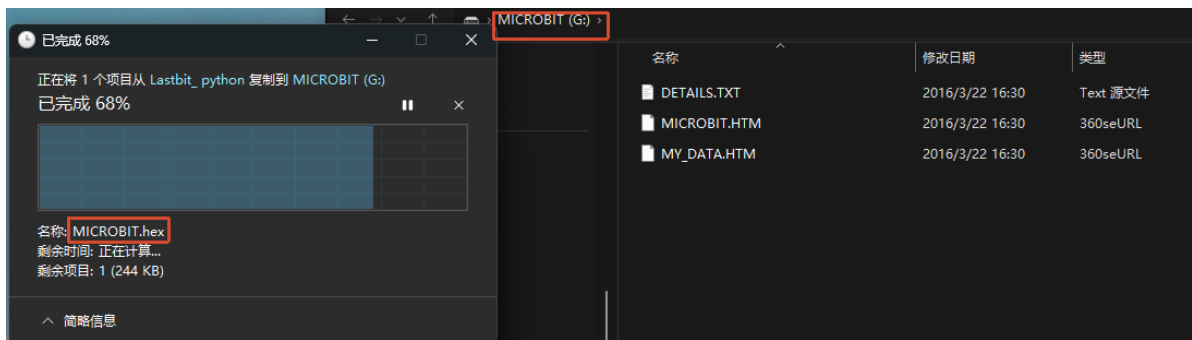
In this section, we will learn to use the accelerometer sensor to detect tilt and achieve the "Ticklish" effect. When the car is detected to be tilted (simulating being tickled), the car will perform a series of interesting actions, including moving up and down, looking left and right, probing with the gripper, shaking the head to refuse, and wiggling with small steps.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



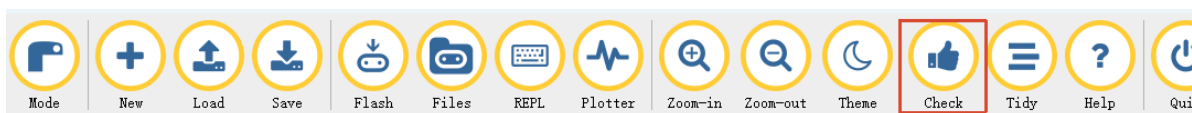
1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.
Click the **New** button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

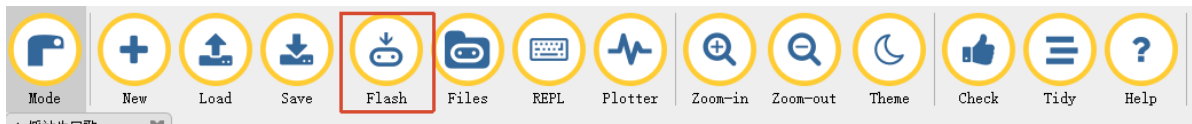
4. Now you can click **Check** to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows **Copied code onto micro:bit.** at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 8. Ticklish Little Guy
from microbit import *
from lastbit import LASTBIT
import music

Lastbit = LASTBIT()

# Speed
speed = 100
# Delay time
delay_time = 200

# Action number
act = 0

# Initialize servos
Lastbit.set_all_servos(90)
sleep(1000)

# Action 0: Up and down
def action_up_down():
    Lastbit.set_servo('S1', 90)
    Lastbit.set_servo('S2', 90)
    sleep(300)

    for _ in range(3):
        # Rise + close claw
        Lastbit.set_servo('S2', 150)
        Lastbit.set_servo('S1', 180)
        sleep(200)

        # Lower + open claw
        Lastbit.set_servo('S2', 90)
        Lastbit.set_servo('S1', 90)
        sleep(200)

# Action 1: Look around
def action_look_around():
    for _ in range(2):
        Lastbit.set_servo('S3', 150)
        Lastbit.set_servo('S1', 90)
        sleep(300)

        Lastbit.set_servo('S3', 30)
        Lastbit.set_servo('S1', 180)
        sleep(300)
```

```

Lastbit.set_servo('S3', 90)

# Action 2: Try grab
def action_try_grab():
    Lastbit.set_servo('S1', 0)
    Lastbit.set_servo('S3', 90)
    sleep(300)

    Lastbit.set_servo('S1', 180)
    sleep(200)

    Lastbit.set_servo('S1', 90)
    sleep(400)

    Lastbit.set_servo('S1', 180)
    sleep(200)

# Action 3: Shake head no
def action_no_no():
    for _ in range(3):
        Lastbit.set_servo('S3', 150)
        Lastbit.set_servo('S1', 180)
        sleep(150)

        Lastbit.set_servo('S3', 30)
        Lastbit.set_servo('S1', 180)
        sleep(150)

    Lastbit.set_servo('S3', 90)

# Action 4: Wiggle with head
def action_wiggle():
    for _ in range(2):
        Lastbit.set_servo('S3', 150)
        Lastbit.car_control('SPINLEFT', speed, delay_time)
        Lastbit.stop_motor()

        Lastbit.set_servo('S3', 30)
        Lastbit.car_control('SPINRIGHT', speed, delay_time)
        Lastbit.stop_motor()

    Lastbit.set_servo('S3', 90)

# Execute next action
def play_next_action():
    global act
    act = (act + 1) % 5

    if act == 1:
        action_look_around()
    elif act == 2:
        action_try_grab()
    elif act == 3:
        action_no_no()

```

```

elif act == 4:
    action_wiggle()
elif act == 0:
    action_up_down()

while True:
    # Read X-axis acceleration
    accel_x = accelerometer.get_x()

    # Tickled
    if accel_x >= 70:
        display.show(Image.HAPPY)
        Lastbit.set_headlight_both('BLUE')

        music.play(music.POWER_UP, wait=False)

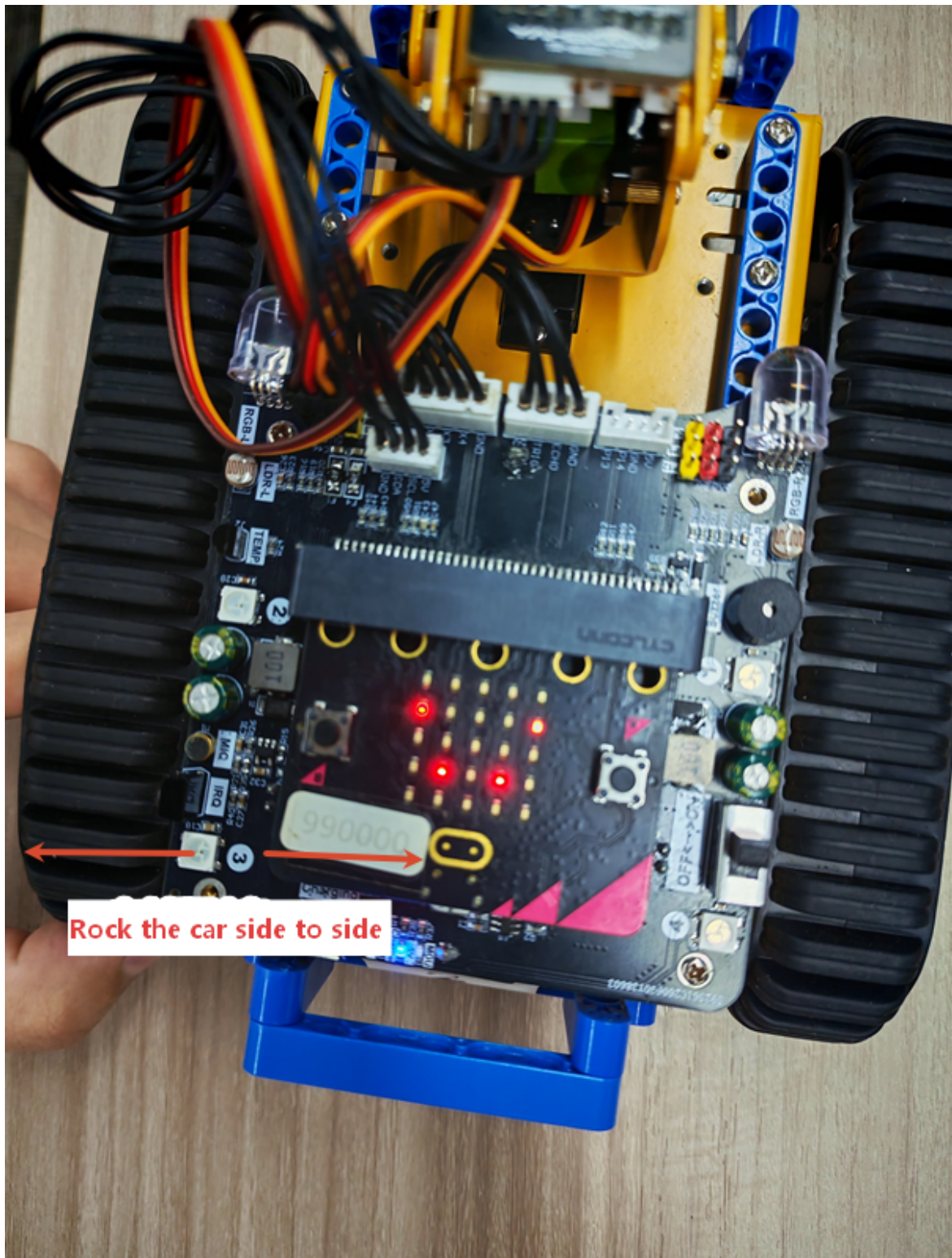
        # Execute an action
        play_next_action()
        Lastbit.stop_motor()

    display.clear()
    Lastbit.all_lights_off()
    sleep(150)

```

Experimental Phenomenon

When the car is detected to be swayed left and right, it will display a happy expression, light up the blue headlights, play a prompt sound, and execute five interesting actions in sequence: moving up and down, looking left and right, probing with the gripper, shaking the head to refuse, and wiggling with small steps. Each time it is gently tilted, the action switches to the next one.



Notes

Make sure the servos are correctly connected and the battery is fully charged to drive the servo movement.